Bryce Rogers

Ph.D., M.S.

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SUMMARY

Lover of Christ and His word, aiming to work in web, product & data-based development for His glory. Ph.D. computational scientist with 4.5 years graduate-level data science, modeling, analysis, visualization, research and development experience (Python & Git ecosystems). Web applications/pages from idea or template to versionable site (UI/UX) in days to weeks. 7 years paid quantitative science and communication

CORE COMPETENCIES

- Languages: *Python, AI-assisted: Bash, R, JavaScript, HTML, CSS*
- AI & Development Tools: IDE & Web-based *AI-assisted backend/frontend development (very competent with GitHub Copilot/Copilot Chat: agentic PRs, local and cloud agents, prompt engineering, model selection, human-guided agentic code review testing CI/CD & documentation)*, VPN/SSH/SFTP, HPC Workloads, Virtual Environments, UNIX/Bash CLI, VPN, GitHub, Fork (Git)
- Modeling & Analysis: Predictive modeling (machine learning, deep learning, and statistical methods; regression, classification), custom neural networks (image & pseudo-speech classification), data cleaning & transformation (e.g. standardization), statistical inference (e.g. uncertainty estimation), data analysis, visualization
- Communication: Coherent technical documentation and presentation across report, slide, or figure; substantive experience communicating scientific or analytical concepts verbally, visually or in writing to technical and non-technical audiences
- Engineering & Solution Cycle: Learn the problem landscape, define your goal, outline the steps toward your goal, implement them, collect your data, interpret it, learn from it

EXPERIENCE

Independent Developer | *USA* | 2025-26 (1.25 years)

- AI-assisted web development *(GitHub Copilot, JavaScript, HTML, CSS, Ruby deps.)*
- Created, developed a *Bible learning game* "Iron: Sharpen Bible Knowledge" (link)
  - Approx. 3 weeks from concept/design images and notes to stable, standalone web version with rich UI/UX (hosted free, GitHub Pages)
  - Full-size/tablet design, plus mobile compatibility features (e.g., window resize, touch-to-hold priority on verse cards)
- Created, developed an *audio visualizer based on human sound perception*, the "Audio Perception Visualizer" (link)
  - 1 week from concept/design images and notes to stable, standalone web version with robust full-size UI/UX (hosted free, GitHub Pages)
  - Flex design (full-size, tablet & ultimately mobile-targeted)
- Leveraged, developed *portfolio website* (link)
  - < 1 week from Jekyll template to clean, responsive site. Custom backend & modals

Graduate Research Assistant | *Kevin Brown Lab, Oregon State University* | 2020–25 (4.5 years)

- Developed and characterized deterministic and neural-network models of word recognition and vision *(Python, Git ecosystems, hybrid/remote work)*
- First known recurrent, predictive coding neural-network classifier of over-time, speech-like input with benchmark behavioral realism in human word recognition (first-author manuscript in 2026, *Neural Computation*, in press)

Graduate Research Assistant | *Brian McNaughton Lab, Colorado State University* | 2013–16 (2.5 years)

- Graduate-level quantitative molecular biology. Assisted in characterizing the binding interaction between a designer and oncoprotein (second-author manuscript in 2014, *Biochemistry*)

FINANCIAL AWARDS

- > $105K earned in personal academic financial awards outside of graduate assistantships. *OSU:* CoE Dean's Award (2022), ARCS Foundation Scholar (2020), Provost Distinguished Graduate Scholar (2020); *UPS:* Trustee Scholarship (2008)

EDUCATION

Ph.D. Bioengineering (Biological Data Sciences Minor) | *Oregon State University* | 2020–25 | GPA: 3.95

- Computational modeling, data science & machine learning in biology
- Data science, programming and analysis-intensive courses: Programming & Data Structures, Applied Machine Learning, Command Line Data Analysis (UNIX/Bash), Intro to UNIX/LINUX, Data Science Tools & Programming, Modern Statistical Methods for Large & Complex Datasets, Foundations of Data Analytics (Python, R, Bash)

M.S. Chemistry | *Colorado State University* | 2013–16 | *Ph.D. Candidacy* | GPA: 3.52

B.S. Biochemistry, Spanish | *University of Puget Sound* | 2008–12 | GPA: 3.05

PUBLICATIONS

- "Extending a formal model of predictive coding to human spoken word recognition." – B.R., M.Y.L., T.H., J.S.M., and K.B. *Neural Computation*, 2026 (in press)
- "Predictive coding models in human spoken word recognition." – B.R. *OSU*, 2025
- "Characterization of the Binding Interaction between the Oncoprotein Gankyrin and a Grafted S6 ATPase." – A.C., B.R., and B.M. *Biochemistry*, 2014

FORMAL COMMUNICATION

- Formal poster presentations:
  - "Predictive Coding Models in Human Cognition." *ARCS Annual Scholar Event* (2022), *Oregon Bioengineering Symposium* (2022)
  - "Formal Predictive Coding Models in Human Vision and Beyond." *Oregon Bioengineering Symposium* (2021)
- Teaching assistant: undergrad. chemistry, biology and eng. (7 sem., 3 qtr. 2013-24)